

New Ability #3: Episodic Memory

In September 1953, a twenty-seven year old man named Henry Molaison underwent an experimental procedure that removed his entire hippocampus—the source of his debilitating seizures. The surgery was, in one sense, a success: severity of his seizures were markedly reduced and he retained his personality and intellect. But his doctors quickly discovered that the surgery had deprived their patient of something precious: upon waking up, Molaison was entirely unable to produce new memories. He could hold a conversation for a minute or two, but shortly thereafter would forget everything that just happened. Even forty years later, he could fill out crosswords with facts from before 1953, but not with facts that occurred after. Molaison was stuck in 1953.

We don't review the past only for the purpose of considering alternative past choices; we also review the past to remember previous life events. People can easily recall what they did five minutes ago or what they majored in at college or that funny joke made during a wedding speech. This form of memory, in which we recall specific past episodes of our lives, is called "episodic memory." This is distinct from, say, procedural memory, where we remember how to do various movements, such as speaking, typing, or throwing a baseball.

But here is the weird thing—we don't truly remember episodic events. The process of episodic remembering is one of simulating an approximate re-creation of the past. When imagining future events, you are simulating a *future* reality; when remembering past events, you are simulating a *past* reality. Both are simulations.

We know this based on two pieces of evidence. First, remembering past events and imagining future events use similar if not the same neural circuitry. The same networks get activated when you're imagining the future as when you're remembering the past. Remembering specific things (faces, houses) reactivates the same neurons in the sensory neocortex as when you actually perceive those things (just as we saw with imagining things).

The second piece of evidence that episodic memory is just a simulation comes from phenomena associated with such memories. For example, it turns out people's episodic memories are "filled in" during the remembering process (much like the filling in of visual shapes). This is why

episodic memories feel so real but are much less accurate than we think. The clearest demonstrations of episodic memory's flaws are in eyewitness testimonies. Someone looking at a lineup of potential criminals is often 100 percent sure which member of the lineup committed the crime. However, contrary to our perceptions of how accurate our memories are, it has been shown that eyewitness testimony is terrible: 77 percent of the wrongfully convicted individuals exonerated by the Innocence Project were originally convicted because of mistaken eyewitness testimony. The distinction between a made-up imagined scene and an actual episodic memory is thin in the neocortex; studies show that repeatedly imagining a past event that did *not* occur falsely increases a person's confidence that the event *did* occur.

One can test the presence of episodic memory in animals by asking them *unexpected* questions about recent events. For example, a rat may learn that when presented with a certain maze, it will find food down one path if and only if it encountered food in the prior few minutes; otherwise, it should go down a different path to get food. The fact that this question is asked randomly (by randomly presenting the rat with this maze) makes it hard to see how the TD-learning mechanisms of the first vertebrates would learn this contingency; the maze is equally paired with each direction. And yet rats easily learn, when placed in this maze, to recall whether they recently encountered food and then choose the right corresponding path to get more food. The only nonmammals that have been demonstrated to have such episodic memory are birds and cephalopods, the two groups of species that seem to have independently evolved their own brain structures for rendering simulations.

After Molaison's surgery, he became the most studied neuroscience patient in history: why was the hippocampus required for creating new episodic memories, but not for retrieving old ones? This is an example of evolution repurposing old structures for new purposes. In mammal brains, episodic memory emerges from a partnership between the older hippocampus and the newer neocortex. The hippocampus can quickly learn patterns, but cannot render a simulation of the world; the neocortex can simulate detailed aspects of the world, but cannot learn new patterns quickly. Episodic memories must be stored rapidly, and thus the hippocampus, designed for the rapid pattern recognition of places, was repurposed to also aid in the rapid encoding of episodic memories.